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JEE MAIN PHYSICS DPP: NEWTON'S THIRD LAW & ACTION-REACTION

Topic: Newton's 3rd Law, Action-Reaction Pairs, Contact Forces & Momentum Conservation | Time: 45 Min |
Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** Action and reaction forces referred to in Newton's third law of motion:
- Act on the same body in opposite directions
 - Act on two different interacting bodies simultaneously
 - Must be equal in magnitude but need not be collinear
 - Act with a finite time delay between action and reaction
- Q2.** A horse pulls a cart along a flat road. The force that primarily accelerates the combined horse-cart system forward is:
- The force exerted by the horse on the cart
 - The force exerted by the cart on the horse
 - The horizontal friction force exerted by the ground on the horse's hooves
 - The internal muscular tension of the horse
- Q3.** A book of mass m rests quietly on a horizontal table. The reaction force to the gravitational force exerted by Earth on the book is:
- The normal contact force exerted by the table on the book
 - The normal contact force exerted by the book on the table
 - The gravitational pull of the table on the book
 - The gravitational force exerted by the book on the Earth
- Q4.** Two blocks of masses $m_1 = 4$ kg and $m_2 = 2$ kg are in contact on a frictionless horizontal floor. A horizontal pushing force $F = 18$ N is applied to m_1 towards m_2 . The magnitude of the contact force between the two blocks is:
- 6 N
 - 12 N
 - 18 N
 - 9 N
- Q5.** In the previous question, if the same force $F = 18$ N is applied to m_2 pushing it towards m_1 , the magnitude of the contact force between the blocks becomes:
- 6 N
 - 12 N
 - 18 N
 - 9 N
- Q6.** A man of mass $M = 60$ kg stands on a spring scale inside an elevator. If the elevator cable snaps and it falls freely under gravity ($g = 10 \text{ m/s}^2$), the normal contact force exerted by the scale on the man is:
- 600 N
 - 1200 N
 - 0 N
 - 300 N
- Q7.** A rocket operates in the vacuum of outer space where there is no surrounding atmosphere. The rocket is propelled forward because:
- Exhaust gases push against the surrounding cosmic dust
 - The atmosphere pulls the rocket from ahead
 - Gravity is absent in space
 - The ejected exhaust gases exert an equal and opposite forward reaction force on the rocket
- Q8.** A body of mass m_1 exerts a force $\vec{F}_{12}$ on another body of mass m_2 . If $\vec{F}_{21}$ is the force exerted by m_2 on m_1 , which of the following is strictly true under all circumstances?
- $\vec{F}_{12} + \vec{F}_{21} = \vec{0}$
 - $|\vec{F}_{12}| > |\vec{F}_{21}|$ if $m_1 > m_2$
 - $\vec{F}_{12} \cdot \vec{F}_{21} > 0$
 - $\vec{F}_{12}$ and $\vec{F}_{21}$ cancel each other and produce zero acceleration
- Q9.** A cannon of mass $M = 500$ kg fires a cannonball of mass $m = 5$ kg horizontally with a muzzle speed of $v = 100$ m/s. The recoil velocity of the cannon is:
- 10 m/s
 - 1.0 m/s
 - 0.1 m/s
 - 5.0 m/s
- Q10.** A swimmer pushes water backwards with his hands and feet to move forward in a pool. The forward force that accelerates the swimmer is:
- The muscular force exerted by the swimmer on the water
 - The buoyant force of water
 - The reaction force exerted by the water on the swimmer
 - The gravitational force on the swimmer

- Q11.** Three identical blocks A , B , and C , each of mass $m = 2$ kg, are connected by light strings and pulled along a frictionless horizontal table with a force $F = 12$ N applied to block C . The tension T_1 in the string connecting A and B is:
- 8 N
 - 6 N
 - 12 N
 - 4 N
- Q12.** In Q11, the tension T_2 in the string connecting block B and block C is:
- 8 N
 - 4 N
 - 6 N
 - 10 N
- Q13.** A person of mass $M = 50$ kg stands on a frictionless ice surface and throws a stone of mass $m = 0.5$ kg horizontally with speed $v = 20$ m/s. The distance moved by the person in 5 seconds after throwing the stone is:
- 0.5 m
 - 1.0 m
 - 2.0 m
 - 0.2 m
- Q14.** When a heavy truck collides head-on with a small compact car:
- The truck exerts a greater magnitude of force on the car than the car exerts on the truck
 - The car exerts a greater magnitude of force on the truck
 - Both vehicles exert equal and opposite forces of identical magnitude on each other
 - The force depends on which vehicle was moving faster
- Q15.** A block of mass m is suspended by a light vertical rope. The action force is the downward pull of Earth on the block (mg). The reaction to the tension force exerted by the rope on the block is:
- The tension force exerted by the ceiling on the rope
 - The gravitational pull of the block on the Earth
 - The normal force on the ceiling
 - The downward force exerted by the block on the rope
- Q16.** A helicopter of mass $M = 2000$ kg hovers stationary in air. It pushes down air at speed $v = 20$ m/s. Taking $g = 10$ m/s², the mass of air pushed down per second ($\frac{dm}{dt}$) is:
- 1000 kg/s
 - 500 kg/s
 - 2000 kg/s
 - 100 kg/s
- Q17.** A person jumps off a small boat onto the shore of a lake. As he steps forward, the boat moves backwards into the water. This observation is a direct consequence of:
- Inertia of rest of the water
 - Newton's third law (Action-reaction pair between foot and boat)
 - Atmospheric pressure gradient
 - Buoyancy force variation
- Q18.** A massive block of mass m_1 rests on top of another block of mass m_2 which rests on a table. The normal contact force exerted by block m_2 on block m_1 and the normal force exerted by block m_1 on block m_2 :
- Form an equilibrium pair on m_1
 - Are unequal if the system accelerates upward
 - Form a valid Newton's third law action-reaction pair
 - Are both equal to $(m_1 + m_2)g$
- Q19.** Two skaters A and B of masses 40 kg and 60 kg face each other on frictionless ice. Skater A pushes skater B with a constant force of 120 N for 0.5 s. The speed acquired by skater A after the push is:
- 1.0 m/s
 - 2.0 m/s
 - 0.5 m/s
 - 1.5 m/s
- Q20.** In Q19, the ratio of the magnitude of acceleration of skater A to that of skater B during the push ($\frac{a_A}{a_B}$) is:
- 3 : 2
 - 2 : 3
 - 1 : 1
 - 9 : 4

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	6	(c)	11	(d)	16	(a)
2	(c)	7	(d)	12	(a)	17	(b)
3	(d)	8	(a)	13	(b)	18	(c)
4	(a)	9	(b)	14	(c)	19	(d)
5	(b)	10	(c)	15	(d)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

Newton's third law explicitly states that forces always occur in matched pairs. If body A exerts a force on body B , then body B simultaneously exerts an equal and opposite force on body A ($\vec{F}_{AB} = -\vec{F}_{BA}$). They act on two different bodies and never cancel each other out.

Sol 2. Correct Option: (c)

To accelerate forward, the horse pushes backward against the ground with its hooves. By Newton's third law, the ground pushes forward on the horse with an equal and opposite horizontal friction force (f_{ground}), which serves as the external force accelerating the system.

Sol 3. Correct Option: (d)

The action-reaction pair to a gravitational force must also be a gravitational force between the same two interacting bodies. The gravitational force of the Earth pulling on the book is paired with the gravitational force of the book pulling on the Earth.

Sol 4. Correct Option: (a)

Common acceleration of the combined system:

$$a = \frac{F}{m_1 + m_2} = \frac{18}{4 + 2} = 3 \text{ m/s}^2$$

The normal contact force N between the blocks accelerates m_2 :

$$N = m_2 a = 2 \times 3 = 6 \text{ N}$$

Sol 5. Correct Option: (b)

When F is applied to m_2 , the acceleration is still $a = 3 \text{ m/s}^2$. The contact force N' now accelerates block m_1 :

$$N' = m_1 a = 4 \times 3 = 12 \text{ N}$$

Sol 6. Correct Option: (c)

In free fall under gravity, the elevator and the man both have downward acceleration $a = g$. Drawing the FBD for the man:

$$Mg - N = Mg \implies N = 0 \text{ N}$$

The scale reads zero (weightlessness).

Sol 7. Correct Option: (d)

The rocket engine expels combustion gases backwards at high velocity. By Newton's third law (and conservation of linear momentum), the expelled gas exerts an equal and opposite forward reaction force (thrust) directly on the rocket body.

Sol 8. Correct Option: (a)

Newton's third law in vector form is $\vec{F}_{12} = -\vec{F}_{21} \implies \vec{F}_{12} + \vec{F}_{21} = \vec{0}$. This holds true universally regardless of the masses, speeds, or motion state of the two interacting objects.

Sol 9. Correct Option: (b)

Initial momentum of the system is zero. By conservation of linear momentum (derived directly from Newton's third law):

$$MV_{\text{recoil}} + mv = 0$$

$$500V_{\text{recoil}} + 500 = 0 \implies V_{\text{recoil}} = -1.0 \text{ m/s}$$

Sol 10. Correct Option: (c)

The swimmer's limbs apply a backward force on the water. By Newton's third law, the water exerts an equal and opposite forward reaction force on the swimmer, which overcomes fluid drag and accelerates the swimmer forward.

Sol 11. Correct Option: (d)

Common acceleration:

$$a = \frac{F}{m_A + m_B + m_C} = \frac{12}{2 + 2 + 2} = 2 \text{ m/s}^2$$

Tension T_1 accelerates only block A :

$$T_1 = m_A a = 2 \times 2 = 4 \text{ N}$$

Sol 12. Correct Option: (a)

Tension T_2 pulls both blocks A and B :

$$T_2 = (m_A + m_B)a = (2 + 2) \times 2 = 8 \text{ N}$$

Sol 13. Correct Option: (b)

By momentum conservation on frictionless ice:

$$Mv_{\text{person}} + mv_{\text{stone}} = 0$$

$$50v_{\text{person}} + 0.5(20) = 0 \implies v_{\text{person}} = -0.2 \text{ m/s}$$

Distance traveled in $t = 5$ s:

$$s = |v_{\text{person}}| \times t = 0.2 \times 5 = 1.0 \text{ m}$$

Sol 14. Correct Option: (c)

By Newton's third law, the force exerted by the truck on the car is strictly equal in magnitude and opposite in direction to the force exerted by the car on the truck ($|\vec{F}_{\text{truck} \rightarrow \text{car}}| = |\vec{F}_{\text{car} \rightarrow \text{truck}}|$). The car suffers more damage purely because its smaller mass undergoes a vastly greater acceleration ($a = F/m$).

Sol 15. Correct Option: (d)

The tension force on the block is the upward contact pull exerted by the rope on the block. The paired reaction is the downward force exerted by the block on the rope.

Sol 16. Correct Option: (a)

For the helicopter to hover in static equilibrium:

$$F_{\text{thrust}} = Mg$$

Thrust force from pushing air down at speed v :

$$F_{\text{thrust}} = v \frac{dm}{dt}$$

$$20 \frac{dm}{dt} = 20000 \implies \frac{dm}{dt} = 1000 \text{ kg/s}$$

Sol 17. Correct Option: (b)

To propel himself onto the shore, the jumper exerts a backward horizontal contact force on the boat with his feet. The boat exerts an equal forward force on the jumper, pushing the boat backwards into the water.

Sol 18. Correct Option: (c)

The normal force exerted by m_2 on m_1 ($\vec{N}_{21}$) and the normal force exerted by m_1 on m_2 ($\vec{N}_{12}$) act on two different bodies across their contact boundary, are equal in magnitude, and are opposite in direction, forming an action-reaction pair.

Sol 19. Correct Option: (d)

By Newton's third law, the force exerted on skater A by skater B during the push is also $F = 120$ N. Acceleration of A:

$$a_A = \frac{F}{m_A} = \frac{120}{40} = 3 \text{ m/s}^2$$

Speed acquired after $t = 0.5$ s:

$$v_A = a_A t = 3 \times 0.5 = 1.5 \text{ m/s}$$

Sol 20. Correct Option: (a)

Since $|F_A| = |F_B| = F$ (action-reaction pair):

$$a_A = \frac{F}{m_A}, \quad a_B = \frac{F}{m_B}$$

$$\frac{a_A}{a_B} = \frac{m_B}{m_A} = \frac{60}{40} = \frac{3}{2}$$

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